INCH-POUND
28 April 1989
MIL-B-87986

MILITARY SPECIFICATION

BEAD BREAKER, PNEUMATIC TIRE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pneumatic tire bead breaker which is rated at ten tons. It is hydraulic in operation and powered by an explosion proof electric motor. It is used to break the head of divided wheel pneumatic tires ranging in size from 15 inches outside diameter (OD) to 58 inches OD and from 4 inches wide to 22 inches wide using ring adapters.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-5041	Tires, Pneumatic, Aircraft
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-B-26195	Box, Wood-Cleated, Skidded, Load-Bearing Base

STANDARDS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addresed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910
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FED-STD-595	Color (Requirements for Individual Color Chips (3 x 5 Supplements) (should be Submitted to the Naval Publications and Forms Center, Philadelphia, Citing FED-STD-595 together with Appropriate Chip Number as shown therein).

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-808	Finish, Materials and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-2073/1A	DOD Material Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

AIR FORCE DRAWING

7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951	Packaging, Commercial.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70	National Electrical Code.

(Application for copies should be addressed to NFPA, Batterymarch Park, Quincy, MA 02269.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Pre-existing design. The bead breaker shall be of a pre-existing design and manufacture having demonstratable satisfactory performance, reliability, and durability. Minor changes to the design will be considered by the procuring activity provided that significant performance, reliability, durability and safety characteristics of the design are retained.

3.3 Design. The bead breaker described herein shall provide a safe, convenient, reliable, and low maintenance means of breaking, or separating, the beads of tires (see 3.3.1) from divided wheel rims using ring adapters. The bead breaker shall be operable by one person. The wheel shall be in a vertical position, between a breaking cylinder and a stationary ram. The space between the cylinder and ram shall be adjustable to accomodate the specified tire widths. An elevating cylinder shall vertically adjust the breaking cylinder and stationary ram. The bead breaker shall be suitable for its intended use. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the bead breaker or any of its components.

3.3.1 Tire and wheel asemblies. The bead breaker shall accommodate Types III, VI, VII, and VIII pneumatic aircraft tires in accordance with MIL-T-5041, ranging in size from 15 inches OD to 58 inches OD and from 4 inches wide to 22 inches wide. The wheel rim diameters range from 6 inches to 32.5 inches.

3.3.2 Temperature range. The bead breaker shall be capable of operation over a temperature range of 0 degrees (Deg) Fahrenheit (F) to +125 Deg F.

3.3.3 Weight. The shipping weight of the bead breaker shall not exceed 2000 pounds.

3.4 Construction. The bead breaker shall be constructed of new parts. Structural elements shall be rigid and possess the requisite strength. Individual components shall be durable and reliable. The bead breaker shall be

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free of defects affecting performance, strength, durability, safety, and reliability. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.4.1 Workmanship. Standards of workmanship shall assure that the bead breaker shall have the strength and efficient operating characteristics found in the manufacturers commercial units.

3.4.2 Hardware. Except as otherwise specified herein or authorized by the procuring activity, MS, AN, or MIL hardware shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial hardware such as screws, nuts, bolts, cotter pins, etc., may be used provided they have suitable properties and are replaceable by the MS, AN, or MIL hardware without alteration, and provided the corresponding MS, AN, or MIL part numbers are referenced in the parts list.

3.4.3 Finishes and Protective coatings. Parts and assemblies shall be finished in accordance with MIL-STD-808 as modified by Air Force drawing 7545352. Final topcoat color shall be aircraft gray, no. 16473, per FED-STD-595.

3.4.4 Identification. All parts shall be marked in accordance with MIL-STD-130, when practical. The following special marking shall be applied to the bead breaker:

Capacity: 10 tons

3.4.5 Instructional panel. Brief and easily readable operating and precautionary instructions shall be permanently affixed to the bead breaker in a readily accessible location. The instructions shall be clear, concise, and sufficient to enable operation of the machine without damage to the equipment or injury to operating personnel.

3.4.6 Assembly. The bead breaker shall be furnished fully assembled except that it may be partially disassembled for ease of packaging and transportation. No welding shall be required for assembly.

3.4.7 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit.

3.5 Components. The bead breaker shall consist essentially of the components described in section 3.5.1 through 3.5.5. Other components shall be furnished as required for a complete, safe, functional bead breaker.

3.5.1 Frame. The frame shall support all components.

3.5.2 Hydraulic system. The hydraulic system shall include as a minimum an electrically powered hydraulic pump, two hydraulic cylinders, a reservoir and necessary check valves, controls, tubing and fittings.

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3.5.2.1 Hydraulic fluid. The hydraulic system shall operate with MIL-H-5606 hydraulic fluid.

3.5.2.2 Pump. The hydraulic pump shall supply fluid to the system at specified operating pressure when powered by an electric motor.

3.5.2.3 Bypass valve. The pump shall have a bypass valve set to bypass fluid from the pump to the reservoir when the hydraulic pressure equals 110 percent of the operating pressure.

3.5.2.4 Cylinders. The breaking cylinder shall be capable of exerting a 10 ton force. The elevating cylinder shall adjust the breaking cylinder and stationary ram over a minimum range of 17 inches to 29 inches above the ground. Both cylinders shall be two way activating.

3.5.2.5 Reservoir. The reservoir shall have a fluid capacity sufficient to fully extend the rams plus a 10 percent reserve.

3.5.3 Electrical system. All electrical equipment and wiring shall conform to the National Electrical Code, NFPA 70. A Class I, Group D, Underwriters Laboratories listed explosion proof electric motor designed and constructed for connection to a source of 220/440 volts, 3 phase, 60 hertz electric power shall be provided, unless otherwise specified (see 6.2).

3.5.4 Controls. The cylinder control levers shall be dead-man type and shall move the same direction as the linear actuators. Operating and service controls shall be readily accessible to an operator wearing heavy gloves. All controls shall be plainly and permanently identified.

3.5.5 Ring adapters. The bead breaker shall be furnished with the minimum number of ring adapters necessary to accommodate the specified tire and wheel assemblies (see 3.3.1). The rings shall be easily installed and and removed. There shall be permanent, protective rubber pads wherever the adapter rings contact the wheels. There shall be no metal contact between any part of the bead breaker and the wheel. The rubber pads shall be located so as to preclude damage to the wheel during bead breaking.

3.6 Performance. The bead breaker shall break both beads on the specified tires (see 3.3.1) with one activation of the breaking cylinder. It shall not damage either the tire or the wheel during the breaking operation. No lifting of the tire and wheel assembly by the operator or the bead breaker shall be required for proper positioning of the ring adapters around the wheel rims. All movements and controls shall be smooth and non-erratic. The bead breaker shall operate as described below without any structural, mechanical, or hydraulic damage or failure.

3.6.1 Leakage. The hydraulic system shall not leak when subjected to a pressure of 1.5 times the system operating pressure for five minutes.

3.6.2 Static proof load. The bead breaker shall be capable of exerting a 15 ton force between the breaking cylinder and stationary ram.

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3.6.3 Reliability. The bead breaker shall perform satisfactorily during and after undergoing 1000 cycles of applying a 10 ton load between the breaking cylinder and the stationary ram.

3.7 Maintainability. The bead breaker shall be designed to permit easy assembly, disassembly, trouble-shooting and maintenance by personnel with a minimum of training and no special tools.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet the requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Test supplies. The contractor shall furnish all necessary facilities, utilities, (for example: electric power) and sample tire and wheel assemblies to demonstrate the bead breaker operation.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.4 First article inspection.

4.4.1 Test sample. First article inspection shall be applied to the first article unit (see 3.1). The first article unit shall be representative of production units the contractor intends to furnish.

4.4.2 Testing. First article inspection shall consist of all tests under 4.6. The tests shall be performed in the order presented. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. Any leakage of hydraulic fluid, evidence of wear, structural damage, or unsatisfactory performance shall be cause for rejection.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, or commercial as specified (see 6.2).

5.1.1 Level A. Level A preservation and packaging shall be in accordance with MIL-STD-2073.

5.1.1.1 Cleaning. Bead breakers shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Bead breakers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservation of component parts shall be in accordance with MIL-P-116.

5.1.1.4 Unit packs. Each bead breaker shall be packaged in accordance with method I of MIL-P-116.

5.1.2 Commercial. The commercial preservation of bead breakers shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Level A packing shall be in accordance with MIL-STD-2073. The bead breakers shall be packed in wood containers conforming to MIL-B-26195, type II, overseas, unless otherwise specified.

5.2.2 Commercial. The commercial preservation of the bead breakers shall be packed in accordance with ASTM D 3951.

5.3 Marking. Marking shall be level A or commercial as specified (see 6.2).

5.3.1 Level A. In addition to any special or other identification marking required by the contract (see 6.2) each exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The bead breaker covered by this specification will be used to break the bead on divided wheel, pneumatic aircraft tires.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this purchase description.

b. Power supply characteristics, if different (see 3.5.3).

c. Additional performance testing, if required (see 4.6.6).

d. Level of preservation and packing and marking (see section 5).

6.3 First article. When a first article inspection is required, the item may be a production item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Bead breaker, pneumatic tire
Divided Wheel, pneumatic aircraft tires
Tire, pneumatic, bead breaker

Custodians:
Navy - YD
Air Force - 99
DLA -CS

Reviewing activity:
Army - AL

Preparing activity:
Air Force - 82

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